

KANAGULK, Australia

WMO: 958270

Lat: 37.1169S

Lon: 141.8031E

Elev: 189

StdP: 99.07

Time Zone: 10.00 (AUS)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.7	1.9	-3.4	2.9	29.6	-1.2	3.5	20.7	10.3	10.8	9.3	10.7	0.7	260	0.546

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.1	37.0	17.7	34.3	17.3	32.0	17.0	20.3	27.6	19.3	27.9	18.4	27.8	5.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.7	21.4	17.0	12.4	20.8	15.7	11.4	20.0	59.4	27.3	55.8	27.7	52.8	27.6	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.4	8.5	7.7	DB	-1.3	41.4	0.9	2.2	-2.0	43.0	-2.6	44.2	-3.1	45.4	-3.8	47.0	(4)
(5)			WB	-1.4	22.0	0.9	1.2	-2.1	22.9	-2.6	23.6	-3.2	24.3	-3.8	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.2	20.9	20.5	18.3	14.7	11.4	9.0	8.5	9.1	10.7	13.2	16.2	18.4	(6)	
	DBStd	5.49	4.24	3.94	3.90	3.30	2.54	1.98	1.89	2.20	2.60	3.61	4.22	4.16	(7)	
	HDD10.0	181	0	0	0	2	13	43	53	43	19	7	1	0	(8)	
	HDD18.3	1828	17	20	49	117	214	281	304	285	229	166	94	50	(9)	
	CDD10.0	1718	339	293	258	144	57	12	7	17	40	106	186	261	(10)	
	CDD18.3	324	97	79	49	9	0	0	0	0	0	7	29	53	(11)	
	CDH23.3	4851	1468	1019	646	135	1	0	0	0	6	167	497	913	(12)	
	CDH26.7	2351	797	496	283	36	0	0	0	0	1	57	224	458	(13)	
(14) Wind	WSAvg	3.7	4.6	4.5	3.9	3.3	3.2	3.0	3.4	3.4	3.5	3.8	4.0	4.1	(14)	
(15) Precipitation	PrecAvg	520	31	26	23	32	43	64	64	67	57	43	32	39	(15)	
	PrecMax	767	135	69	74	89	89	139	112	131	133	86	67	166	(16)	
	PrecMin	302	0	4	2	2	4	11	29	16	23	5	8	5	(17)	
	PrecStd	111	29	17	17	21	23	32	21	30	29	25	16	32	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	37.8	36.4	30.7	22.2	16.4	15.8	19.7	24.0	31.6	35.3	39.6	(19)	
		MCWB	18.8	19.5	17.5	16.7	13.9	12.0	11.9	12.7	14.2	15.5	17.4	17.5	(20)	
	2%	DB	38.0	35.0	32.5	26.9	19.6	14.8	14.0	17.0	20.4	27.9	32.2	35.4	(21)	
		MCWB	17.8	18.1	17.2	15.0	13.0	11.6	11.1	11.5	13.4	14.7	16.7	16.7	(22)	
	5%	DB	34.4	32.5	29.9	24.3	17.5	13.8	12.9	14.8	17.9	24.4	29.3	31.8	(23)	
		MCWB	17.6	17.9	16.7	14.2	12.9	11.1	10.6	11.0	12.5	14.2	16.5	16.2	(24)	
	10%	DB	31.1	29.7	26.8	21.6	15.9	12.8	12.0	13.3	16.0	21.3	25.7	28.5	(25)	
		MCWB	16.8	17.0	16.1	13.6	12.4	10.6	10.2	10.4	11.7	13.7	15.3	15.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.7	21.7	20.1	17.7	15.7	13.6	13.2	13.5	15.6	18.1	20.0	20.7	(27)	
		MCDB	28.6	31.1	26.0	24.8	17.8	14.8	14.2	17.3	20.8	24.9	29.1	26.1	(28)	
	2%	WB	20.5	20.0	18.8	16.5	14.5	12.5	12.0	12.2	14.2	16.2	18.5	18.7	(29)	
		MCDB	28.3	27.0	26.6	22.3	17.1	13.8	12.9	15.4	18.6	22.9	26.8	26.8	(30)	
	5%	WB	19.2	19.1	17.9	15.5	13.7	11.6	11.1	11.4	13.1	15.1	17.4	17.7	(31)	
		MCDB	29.8	27.7	26.1	20.7	16.2	12.9	12.4	14.0	17.0	22.0	25.0	27.3	(32)	
	10%	WB	18.1	18.0	16.9	14.6	12.9	11.0	10.5	10.7	12.0	14.0	16.2	16.7	(33)	
		MCDB	29.4	27.9	25.3	19.8	15.3	12.3	11.7	12.9	15.7	20.8	24.2	26.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	16.1	14.7	13.3	11.8	8.7	8.0	7.2	8.5	9.9	12.8	14.0	15.4	(35)	
		MCDBR	21.8	19.5	18.4	16.8	11.2	9.5	8.1	11.1	13.6	19.1	20.4	21.8	(36)	
	5% WB	MCWBR	5.9	5.8	6.1	6.9	6.1	6.5	5.7	6.6	7.5	8.2	6.9	6.3	(37)	
		MCWBR	16.1	14.9	14.8	12.2	8.7	7.7	7.1	9.7	12.2	15.2	16.2	16.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.338	0.332	0.316	0.307	0.299	0.294	0.295	0.305	0.324	0.336	0.334	0.332	(40)	
		taud	2.497	2.526	2.569	2.577	2.579	2.581	2.577	2.531	2.466	2.440	2.480	2.506	(41)	
	Ebn at Noon		996	978	956	906	853	828	850	892	931	963	993	1006	(42)	
		Edh at Noon	113	105	93	81	70	65	69	83	102	114	115	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.48	6.57	5.16	3.61	2.42	1.95	2.11	2.98	4.13	5.48	6.51	7.27	(44)	
	RadStd		0.43	0.32	0.24	0.22	0.12	0.15	0.12	0.17	0.31	0.44	0.45	0.41	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (27 neighbors)		+0.39	N/A	N/A	+0.89	N/A	N/A	N/A	-96	+132	+39

Nomenclature: See separate page